

EXPLORATORY DATA ANALYSIS · 10 GLOBAL MARKETS

What Actually Drives Airbnb Listing Prices?

A currency-safe analysis of 279,712 listings across Paris, New York, Sydney, Rome, Rio de Janeiro, Istanbul, Mexico City, Bangkok, Cape Town and Hong Kong.

Source: Airbnb Listings & Reviews Dataset (Kaggle) | 33 raw features | Prepared for Data Science & ML review

Project Objective

The dataset has no target variable. Price is the natural analytical focus — but prices are recorded in ten different local currencies, so the first job is to make price comparable at all.

279,712

Listings analysed

10

Cities · 10 currencies

33

Raw features

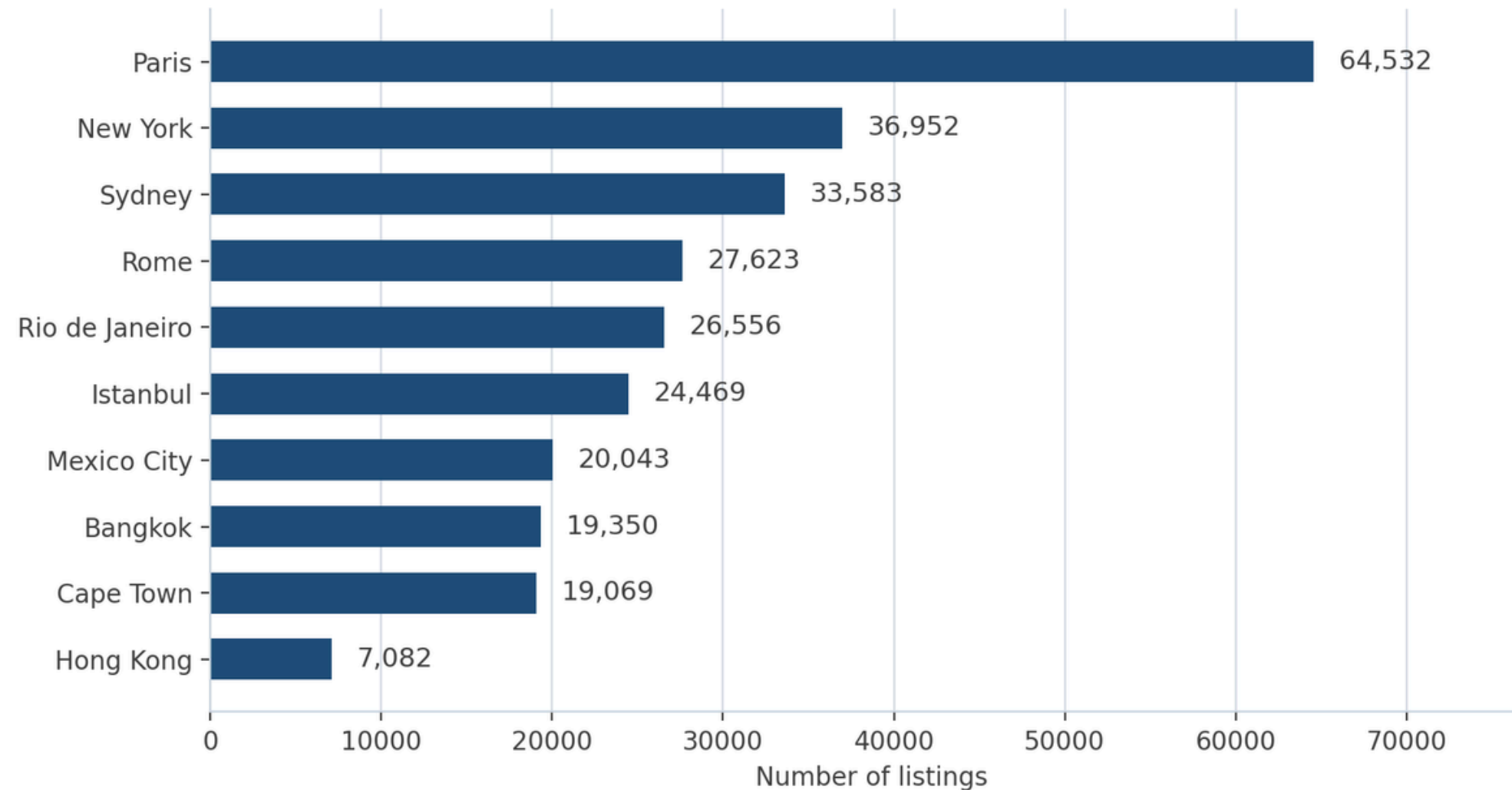
181,647

Unique hosts

- Build a defensible within-city price index so listings can be compared across markets
- Quantify how capacity, room type, amenities, hosts, reviews and location relate to price
- Produce a clean, leakage-aware feature set for the modelling stage

Objective: explain within-city price variation, not raw price levels.

Dataset Overview

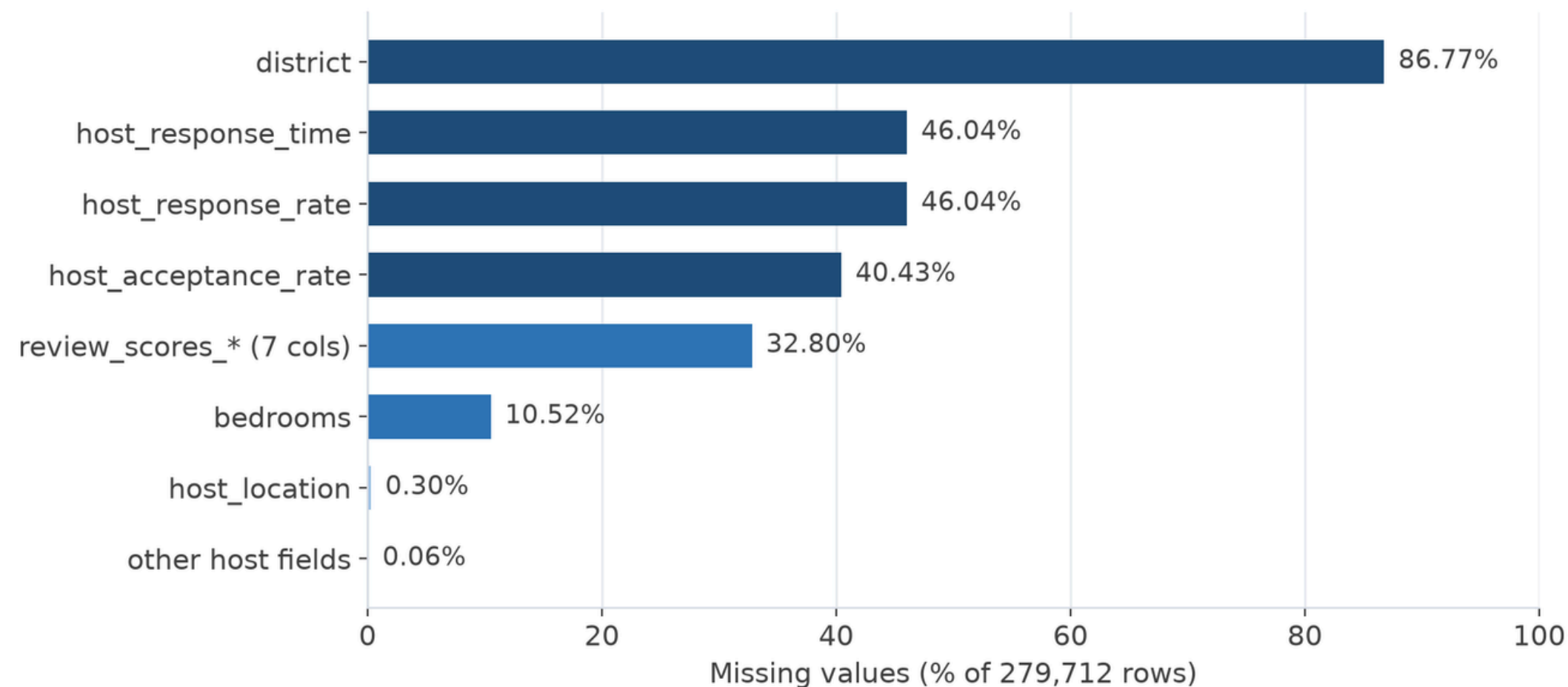


Composition

- 17 numerical · 16 categorical features
- Entire places dominate at 65.1%
- Hotel + shared rooms under 4% combined
- 144 raw property types collapsed into 5 groups
- Paris is the largest market (23.1%), Hong Kong the smallest (2.5%)
- Median listing: 2 guests, 1 bedroom, 2-night minimum

A platform of small, whole-property rentals — not hotel-style inventory.

Data Quality Summary



Treatment decisions

- district (86.8%) and host_location dropped — location already covered
- Host-response and review-score gaps kept unimputed: missingness is systematic, not random
- bedrooms imputed by property-type median
- <1% missing → rows removed; 0 duplicate listings
- 199 invalid rows removed (price ≤ 0 , accommodates ≤ 0 , max < min nights)

~33% of listings have no review history at all — a signal to model, not a gap to impute.

Research Questions

Q1

Can prices be compared across cities?

Ten currencies, no documented FX rates for the collection period.

Q2

Does capacity or room type drive price?

Test structural attributes against a within-city price index.

Q3

Do amenities command a premium?

Individual flags and total amenity count vs price.

Q4

Do Superhosts and pro hosts price higher?

Host status, tenure and portfolio size effects.

Q5

Do better reviews mean higher prices?

Seven review dimensions on the reviewed subset.

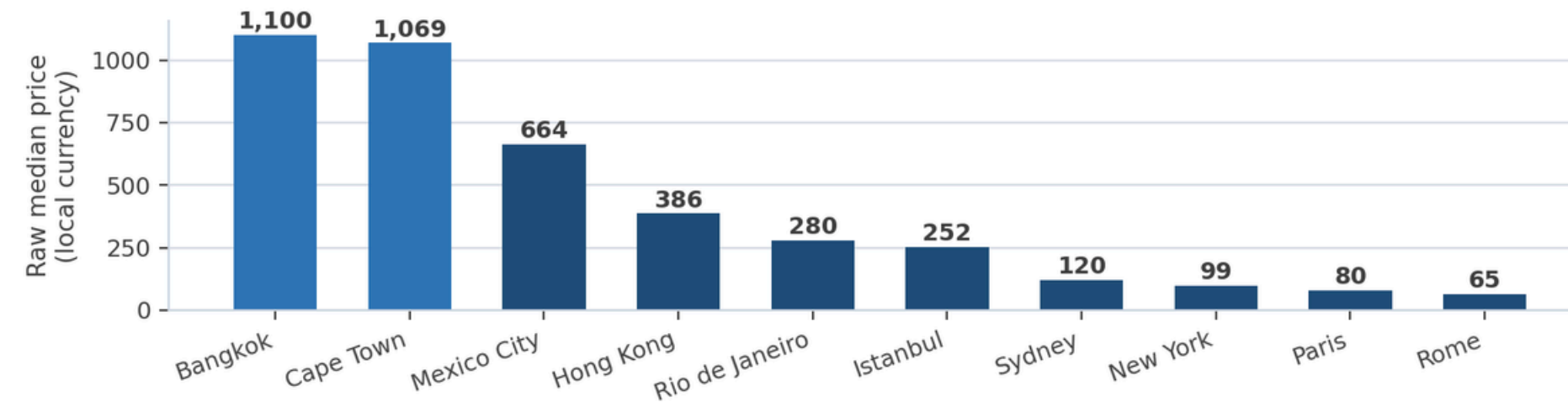
Q6

How much of price is location?

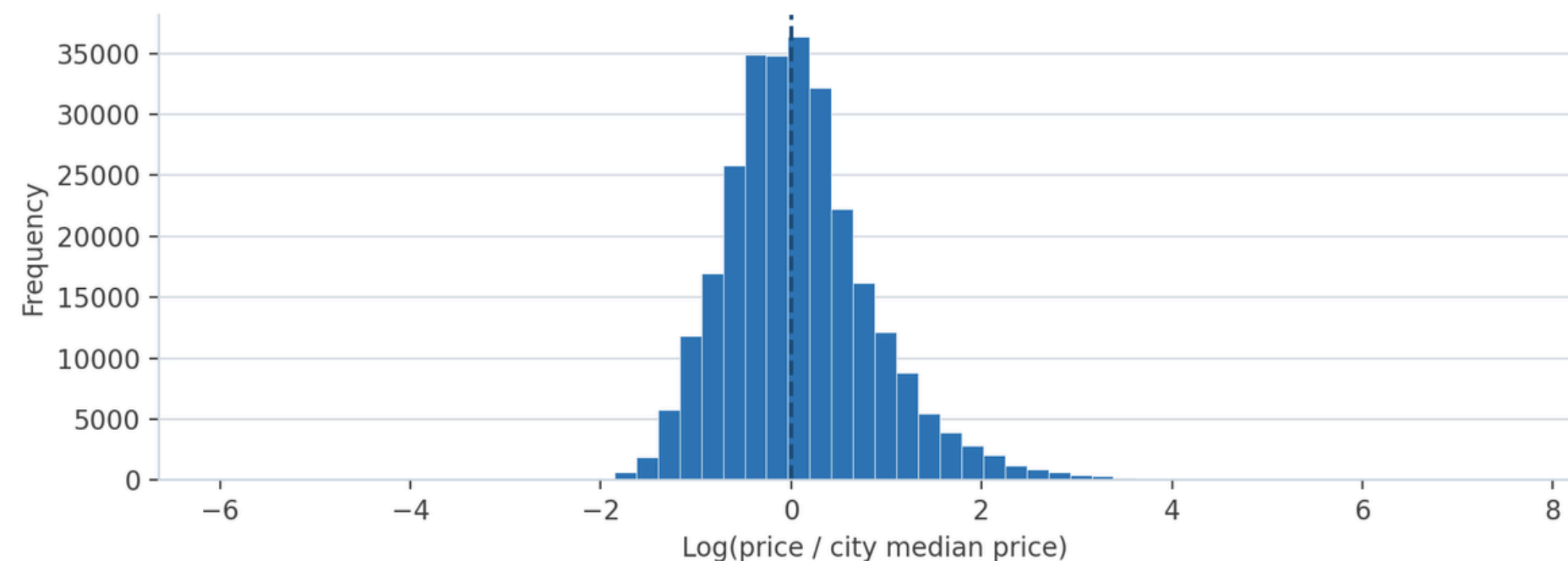
Neighbourhood-level premiums inside each city.

Making Price Comparable

BEFORE — raw median price by city (unconverted)



AFTER — within-city price index (log)

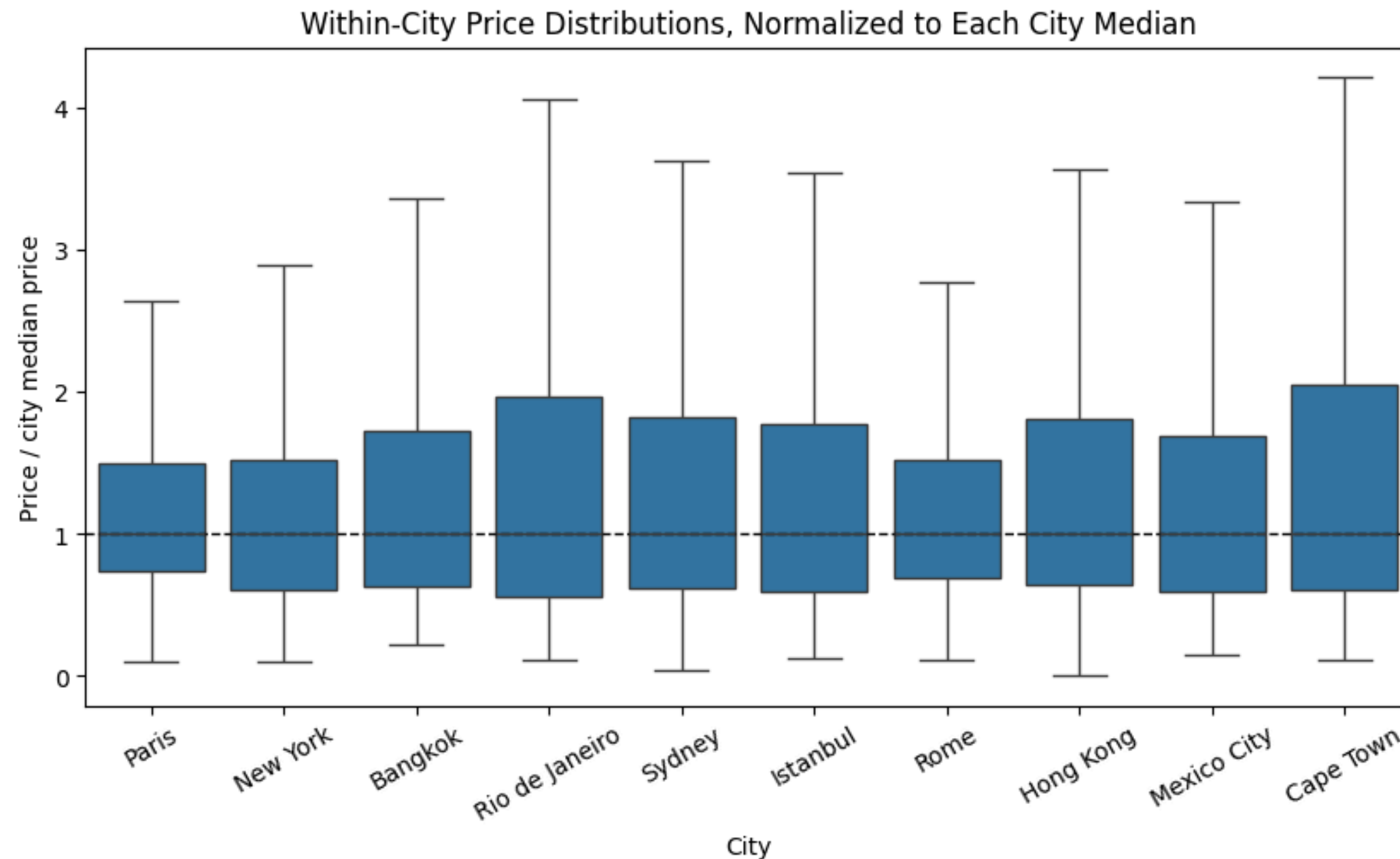


$$\text{price_index_city} = \text{price} \div \text{city median price}$$

- Raw medians ranked Bangkok, Cape Town above Paris — the trigger for this fix
- Log of the index is near-symmetric — suitable for correlation and modelling
- 1.0 = a typical listing in its own market
- Removes the currency problem without inventing exchange rates

Every price finding in this deck is expressed relative to the listing's own city median.

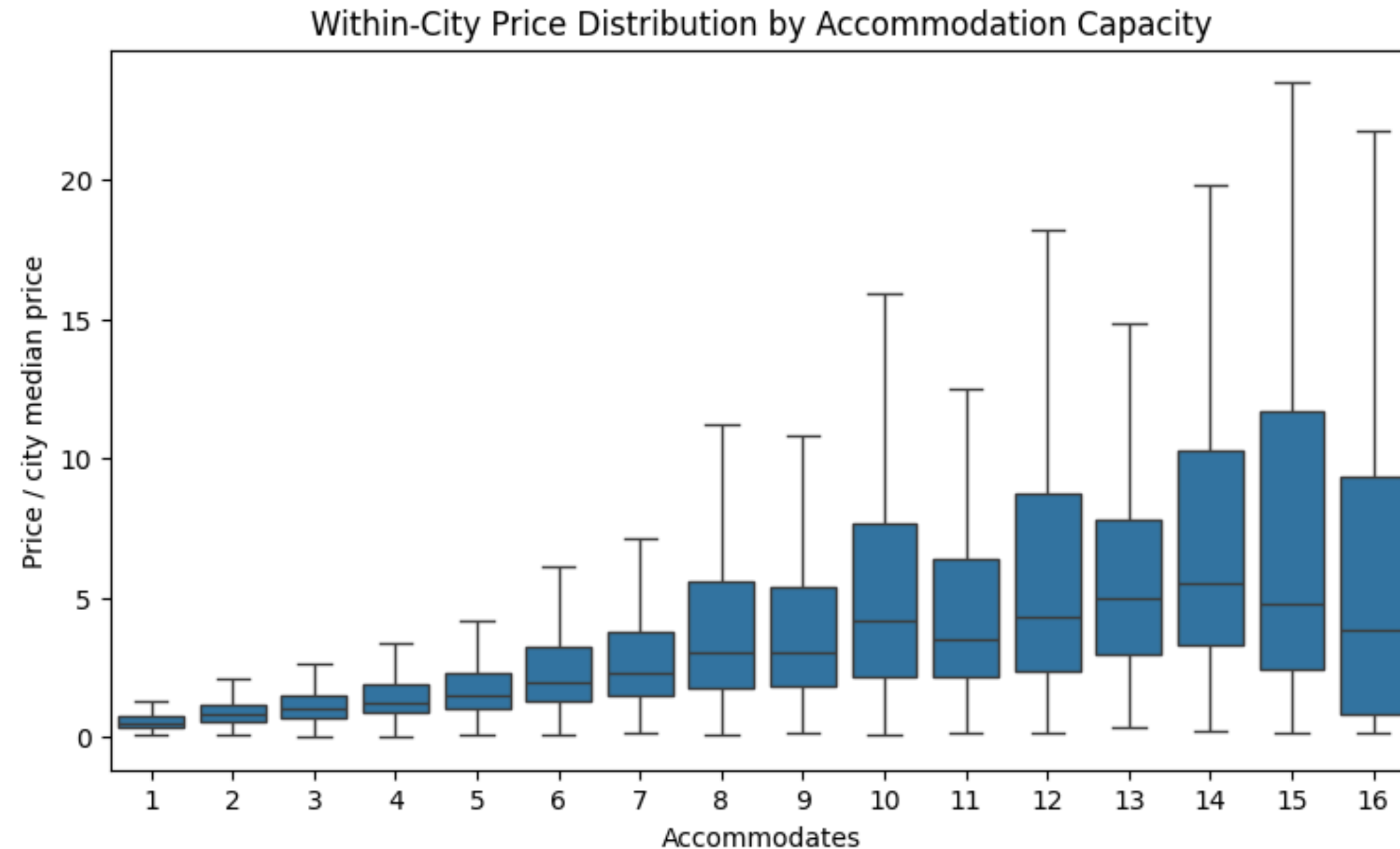
Price Analysis: Spread Within Each Market



- Interquartile ranges are strikingly similar across all ten cities
- Cape Town and Rio show the widest upper spread (2.0x+ of median)
- Paris, Rome and New York are the tightest markets
- Raw medians are NOT ranked — different currencies

Relative price dispersion is a near-universal property of these markets, not a city trait.

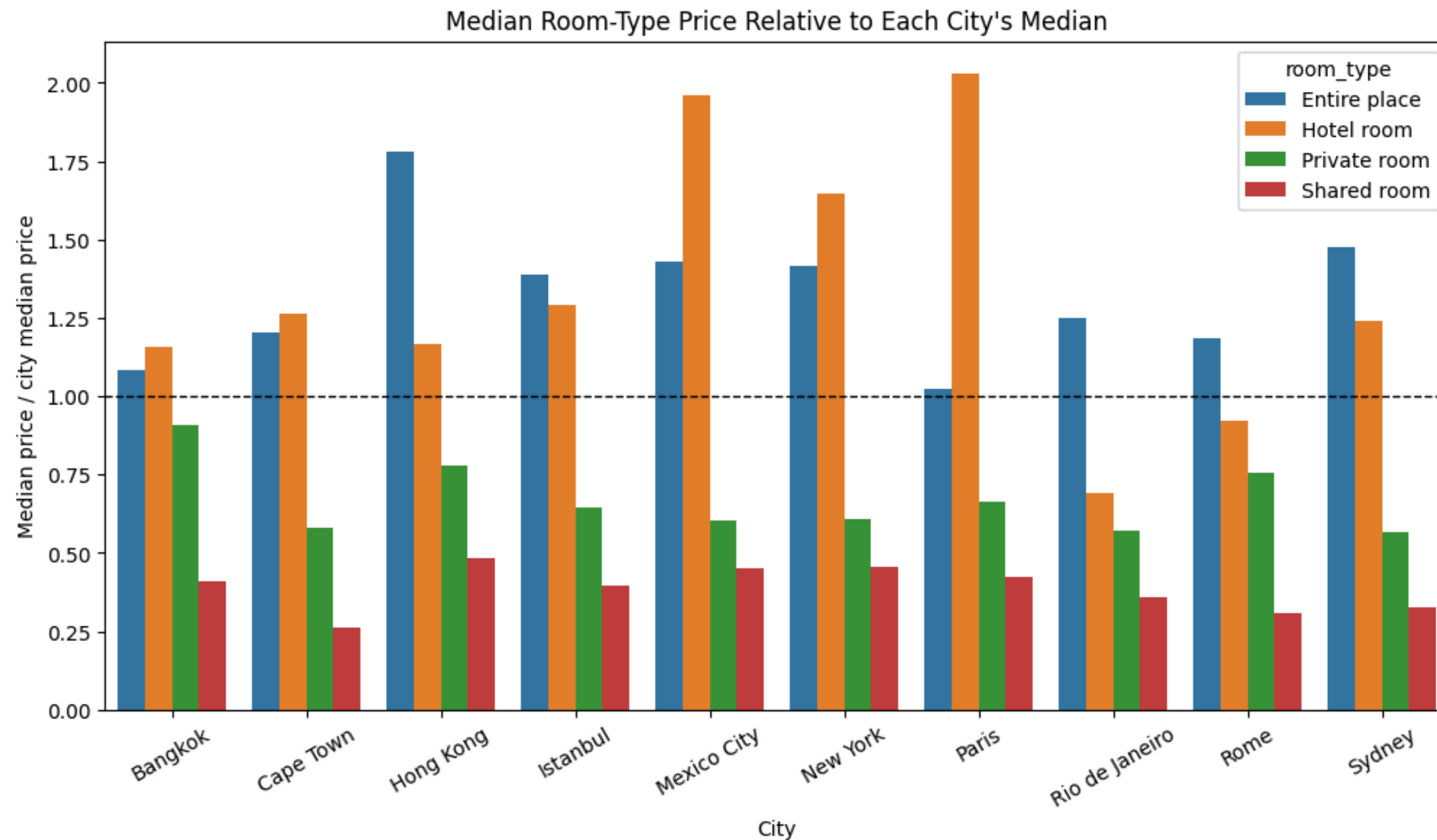
Capacity Is the Strongest Structural Price Driver



- Median price index rises steadily from 0.5x (1 guest) to ~5x (14 guests), with minor dips at the largest, thinly-populated capacities
- Variance grows with capacity — large listings are priced far less consistently
- Median price per guest is only 0.375x — bigger listings are cheaper per head

Price scales with capacity, but sub-linearly: each extra guest adds ~ $\frac{1}{3}$ of a city median.

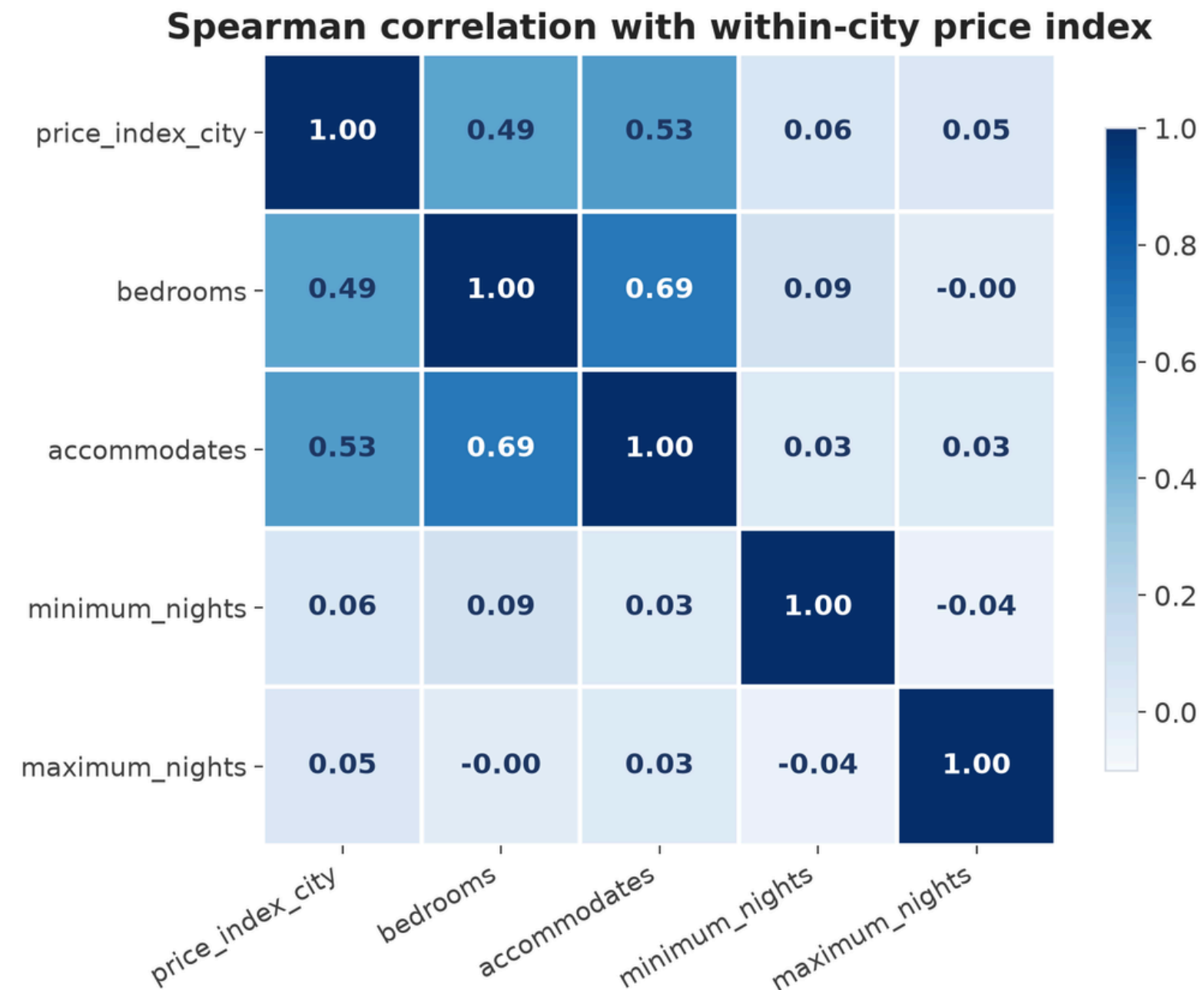
Room Type Positioning Is Consistent Across Markets



- Entire places sit above the city median in all 10 cities
- Private rooms cluster at 0.55x–0.9x, shared rooms at 0.26x–0.49x
- Hotel rooms are the most volatile: 2.03x in Paris vs 0.69x in Rio de Janeiro

Room type behaves like a stable global price tier — a strong categorical feature.

Correlation Structure



Spearman, full cleaned dataset

- accommodates ↔ price index: 0.53 — the strongest single numeric relationship
- bedrooms ↔ price index: 0.49, and 0.69 with accommodates → collinearity to manage
- minimum_nights (0.06) and maximum_nights (0.05) are effectively unrelated to price
- Numeric features alone explain little: price lives in location, type and text

Only size-related features correlate with price; booking constraints carry almost no signal.

Amenities Analysis

0 – 89

Amenity count range (median 18)

≈ 0.15

Spearman: amenity count vs price index

≈ 0.23

Spearman: amenity count vs capacity

121

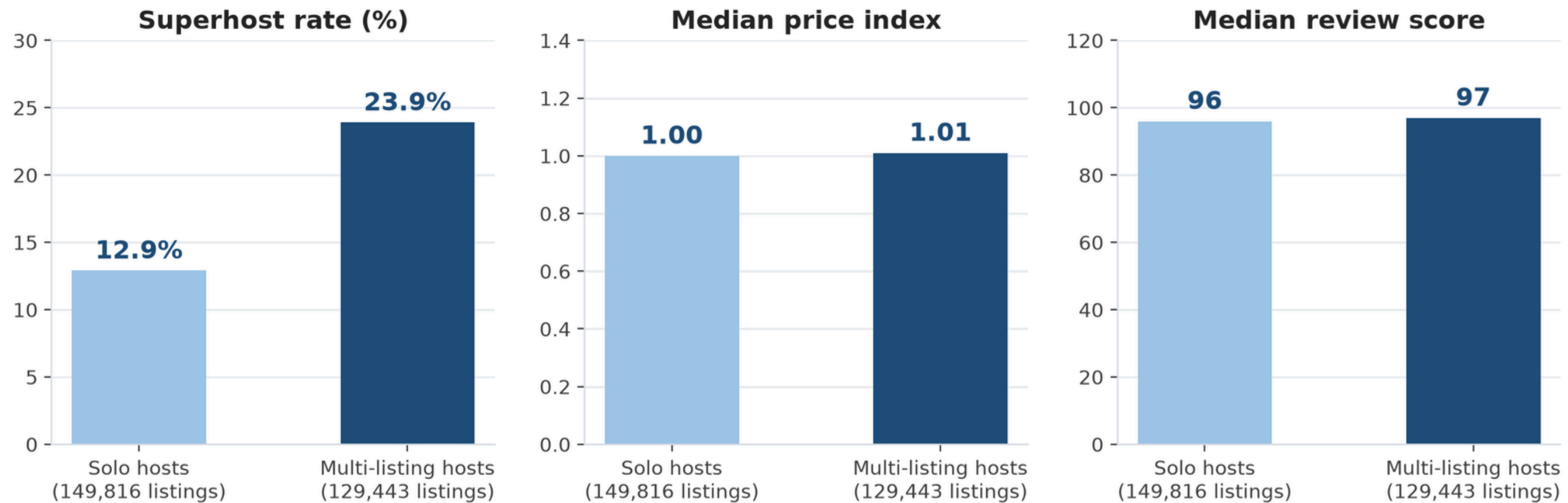
Listings with zero parsed amenities

What the top-20 amenity flags show

- No individual amenity — Wi-Fi, kitchen, pool included — is a strong price driver on its own; all correlations are weak.
- Amenity count tracks capacity more closely than price: larger listings simply list more amenities.
- Naming is unstandardised free text ("Wifi" vs "Pocket wifi"), so amenity features are approximate by construction.

Amenities are a proxy for listing size, not an independent premium.

Host Analysis



- Only 17.99% of hosts hold Superhost status
- Multi-listing hosts are ~2x more likely to be Superhosts (23.9% vs 12.9%)
- Largest host runs 627 Paris listings; 39 hosts exceed 100 listings
- Host tenure vs review score: Spearman ≈ -0.004 (no relationship)

Professional hosts earn status and scale — but not higher prices or better reviews.

Review Analysis

32.7%

Listings with no review scores

-0.02 to +0.01

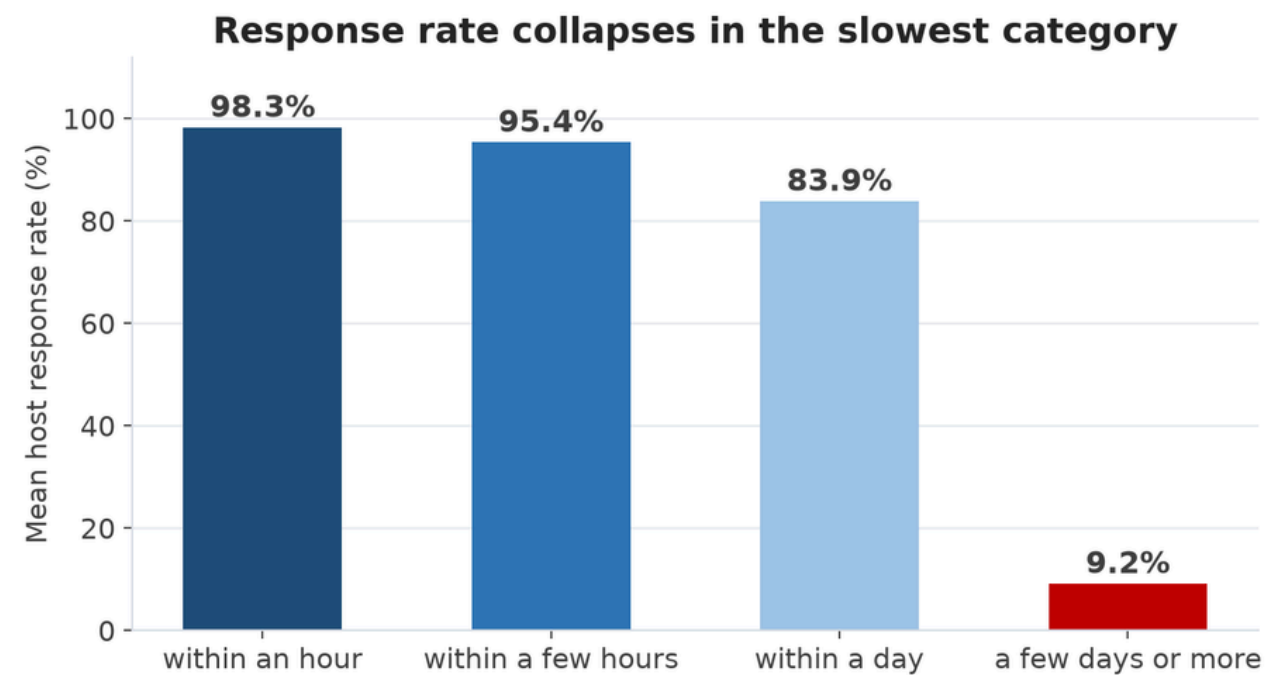
Correlation of all 7 review dimensions with price

1.02 vs 0.96

Superhost vs non-Superhost median price index

-0.004

Host tenure vs review score

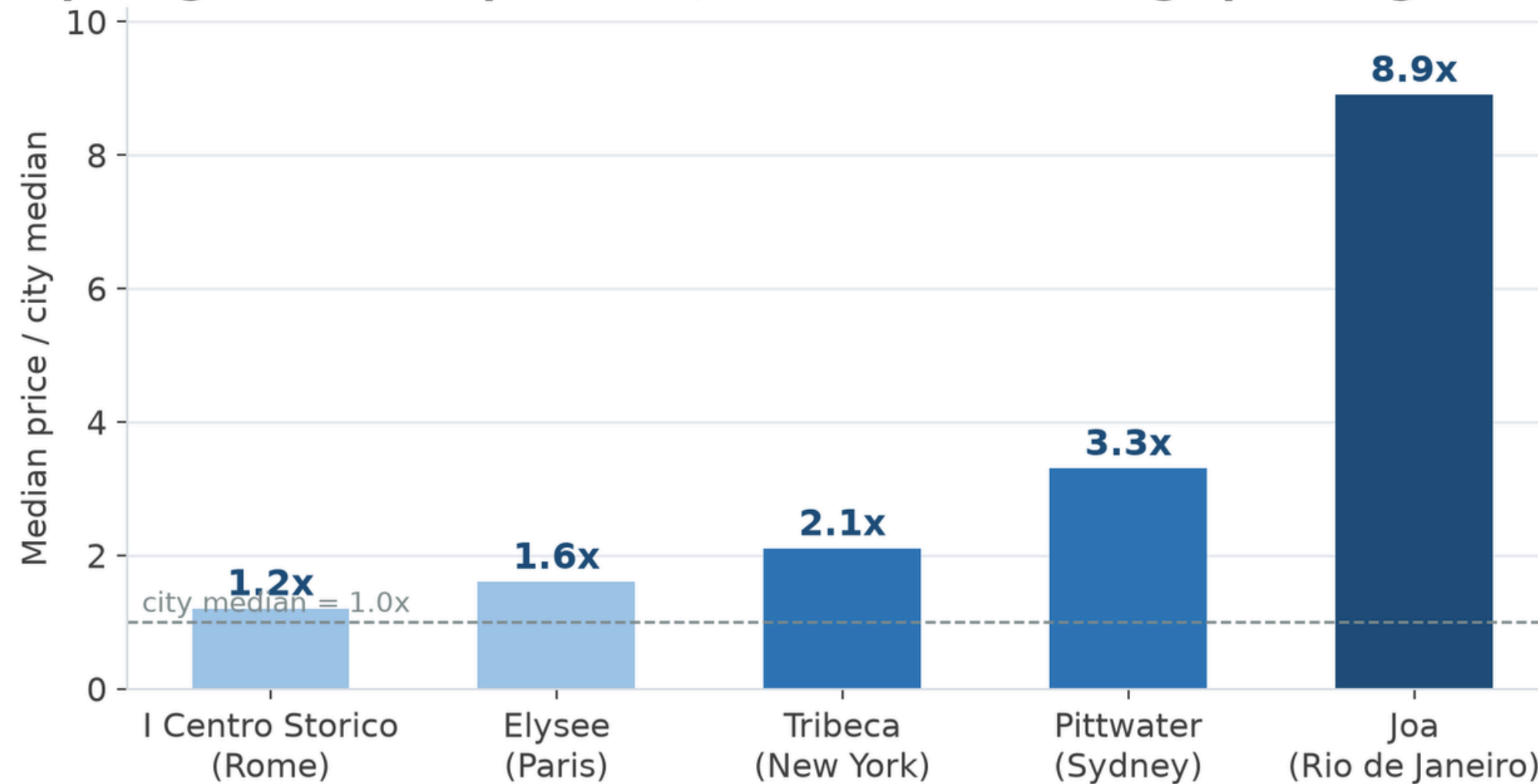


- Response time and response rate measure the same behaviour from two angles
- The slowest bucket collapses to 9.2% — those hosts barely respond at all

Guests do not pay a premium for quality: review scores are effectively price-neutral.

Location Analysis

Top neighbourhood premium, minimum 10 listings per neighbourhood



- Top neighbourhood in most cities sits at 1.2x–2.1x the city median
- Outliers are exclusive enclaves: Joa (Rio) 8.9x, Pittwater (Sydney) 3.3x
- Neighbourhoods under 10 listings excluded to avoid single-listing artefacts
- Geographic scatter shows clear coastal / central clustering in every city

Location is a real within-city price driver — encode distance-to-centre, not just city.

Interaction Analysis

Superhost × Room type

Premium only for Entire place and Hotel room; Superhosts price slightly BELOW peers for private and shared rooms.

Instant bookable × City

Adoption ranges from ~60% in Bangkok to ~30% in New York — a market-specific behaviour, not a platform norm.

Host tenure × Review score

Spearman ≈ -0.004 . Long-tenured hosts do not earn systematically better reviews.

Portfolio size × Price

Median price index 1.00 (solo) vs 1.01 (multi-listing) — scale does not translate into pricing power.

No interaction produces a large effect — pricing power is structural, not behavioural.

Outlier Investigation

3 rows out of 279,712 drove the entire distortion



IQR outliers were kept

- High prices, large properties and long minimum stays are genuine business scenarios, not errors
- The real problem was hidden: 3 rows carried 2,147,483,647 (int32 max) as maximum_nights
- Replacing that sentinel with NaN moved skew from 283.99 to -0.37
- 1,637 listings (0.59%) report more bedrooms than guests — flagged, not removed

A single hidden placeholder in 3 of 279,712 rows fully distorted a summary statistic.

Key Insights

01 Currency discipline is non-negotiable

Raw prices across 10 currencies cannot be pooled; all analysis uses $\text{price} \div \text{city median}$.

03 Quality signals are price-neutral

All 7 review dimensions land between -0.02 and $+0.01$ against price.

05 Missingness is a feature, not noise

~33% of listings have no reviews and 34–46% no host-response history.

02 Size explains price, everything else is weak

accommodates 0.53 and bedrooms 0.49 dominate; booking rules ≈ 0 .

04 Location premiums are real and local

Top neighbourhoods reach 1.2x–8.9x their own city median.

06 Professional hosts scale, not price

23.9% Superhost rate vs 12.9%, yet an identical median price index.

Price is structural: capacity, room type and location — not ratings or amenities.

Limitations

No exchange rates

FX rates for the collection period are undocumented, so absolute cross-city price levels remain out of scope.

Non-random missingness

Review and host-response gaps concentrate in newer listings; naive imputation would bias any model.

Unstandardised amenities

Free-text derived flags ("Wifi" vs "Pocket wifi") make amenity features approximate.

No demand or occupancy data

Price is observed, not realised revenue — booking rates and seasonality are unavailable.

Host concentration

39 hosts hold 100+ listings each, so listing-level rows are not fully independent.

Findings hold within cities and within this snapshot — not as absolute market rankings.

Future Work: Feature Engineering & Modelling

Features already engineered

- `price_index_city · log_price_relative_city`
- `price_per_guest_index`
- `amenity_count · has_wifi / has_pool / has_kitchen`
- `property_type_grouped` (144 → 5 categories)
- `host_tenure_years · has_reviews`

Modelling roadmap

- Target `price_index_city`, not raw price
- Target-encode high-cardinality neighbourhood; one-hot city and room type
- Model `has_reviews` explicitly, or use a two-stage design
- Group-split by `host_id` to prevent leakage across portfolios
- Add distance-to-centre and geospatial cluster features

Next: a currency-safe, leakage-aware gradient-boosted baseline on `price_index_city`.